Integrating Lane/Divider Perception with Speed Sign Recognition for Context-Aware ADAS in Vietnam

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***Abstract*—This paper presents a context-aware perception stack for Advanced Driver Assistance Systems (ADAS) in Vietnam, where speed regulation is often inferred from both explicit signs and infrastructure cues. Instead of treating traffic sign recognition as an isolated detection task, we jointly model (i) speed sign and zone sign detection, (ii) lane and hard-divider segmentation, and (iii) legal-rule inference following Circular 31/2019/TT-BGTVT and QCVN 41:2019/2024. The proposed architecture uses a shared encoder, dual task heads, slicing-aided small-object detection, and temporal tracking to improve robustness under motorcycle occlusion, mixed traffic, non-standard sign placement, glare, and camera vibration from wide-angle dashcams. On a combined dataset built from VNTS, Zalo AI, BDD100K, TT100K, and custom dashcam data, the system reaches 82.6% mAP@0.5 for target signs, 92.3% mIoU for infrastructure segmentation, and 39 FPS on Jetson TX2. Ablation shows that context reasoning and SAHI contribute most to small-object recovery and stable speed-limit estimation under occlusion. These results support deployment of localized, legally grounded ADAS for Vietnamese roads.**

***Index Terms*—ADAS, traffic sign detection, lane segmentation, context-aware inference, Vietnam traffic, multi-task learning**

I. Introduction

Road scenes in Vietnam differ significantly from those assumed by most public ADAS benchmarks. Urban corridors are dominated by motorcycles, traffic flow is heterogeneous, and sign visibility is frequently degraded by roadside objects, buses, and trucks. In addition, practical speed compliance is not always determined by a visible numeric sign; it can also depend on road context, such as entering/leaving populated areas and the presence of hard median dividers. This makes single-task traffic sign recognition (TSR) brittle in real deployment.

Project materials from Group 6, including project introduction and data collection drafts, consistently indicate three failure modes in local settings: domain mismatch from western-centric datasets, small-sign sparsity in wide-angle camera views, and severe occlusion dynamics caused by motorcycles. Therefore, we target a perception system that does not only detect signs but also *infers legal speed context* from multiple visual cues.

II. Related Work

Traditional TSR pipelines based on handcrafted color-shape operators are computationally efficient but unstable under illumination shifts and clutter [1]. Deep object detectors such as Faster R-CNN and YOLO variants improve accuracy-speed tradeoffs [2], [3]. For driving perception, multi-task architectures (e.g., YOLOP) share low-level features across detection and segmentation to reduce compute and improve representation reuse [4].

Recent studies on Vietnamese traffic emphasize non-standard sign placement and small-object difficulty [7], [8]. SAHI-style inference is effective for tiny object recall at high resolution [6]. For joint-task optimization, uncertainty-based loss weighting is widely adopted in autonomous-driving multi-task learning [5]. However, the existing literature seldom integrates legal reasoning rules specific to Vietnam speed policies after visual perception. This paper addresses that gap.

III. Problem Statement

Given a video frame sequence $\{I_t\}_{t=1}^{T}$ from a forward dashcam, we estimate a legal speed state $\hat{v}_t \in \mathcal{V}$ where

$$\mathcal{V} = \{40, 50, 60, 80, 90\} \text{ km/h}. \tag{1}$$

The model predicts object detections $\hat{D}_t$ (speed signs and zone signs) and segmentation masks $\hat{S}_t$ (lane/divider/drivable area):

$$(\hat{D}_t, \hat{S}_t) = f_\theta(I_t). \tag{2}$$

The final speed estimate is produced by legal-context fusion:

$$\hat{v}_t = g\left(\hat{D}_{1:t}, \hat{S}_{1:t}, \mathcal{R}_{\text{VN}}\right), \tag{3}$$

where $\mathcal{R}_{\text{VN}}$ denotes rule constraints from Vietnamese regulations.

IV. Proposed ADAS System Architecture

A. System Pipeline

Fig. 1 illustrates the end-to-end pipeline from camera stream to ADAS speed advisory.

B. Overall Network Design

The perception backbone follows a shared-encoder, multi-decoder scheme (Fig. 2). CSP-style feature extraction and feature pyramid fusion serve both tasks. Detection is optimized for tiny signs, while segmentation preserves boundary details for lane-divider interpretation.

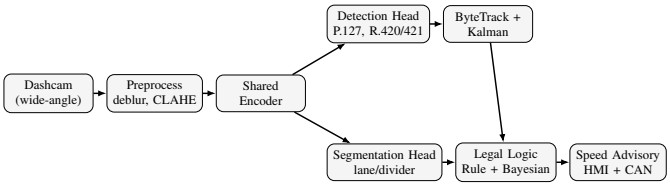


Fig. 1. Context-aware ADAS pipeline for Vietnamese traffic.

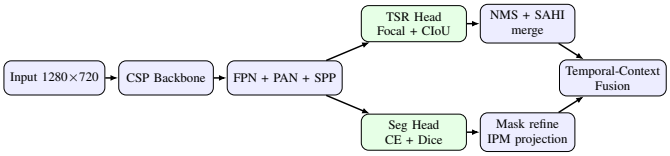


Fig. 2. Overall architecture with shared encoder and dual task heads.

V. Methodology

A. Detection-Segmentation Joint Objective

The full objective is:

$$\mathcal{L} = \lambda_{det}\mathcal{L}_{det} + \lambda_{seg}\mathcal{L}_{seg} + \lambda_{rule}\mathcal{L}_{rule}. \tag{4}$$

Detection combines classification and localization:

$$\mathcal{L}_{det} = \mathcal{L}_{focal} + \beta\mathcal{L}_{CIoU}, \tag{5}$$

$$\mathcal{L}_{focal} = -\alpha_t(1-p_t)^{\gamma}\log(p_t). \tag{6}$$

Segmentation uses weighted CE and Dice losses:

$$\mathcal{L}_{seg} = \mathcal{L}_{CE} + \eta\mathcal{L}_{Dice}. \tag{7}$$

The rule-consistency term penalizes conflicts between perceived context and inferred speed class:

$$\mathcal{L}_{rule} = \sum_{k} \mathrm{CE}\big(y_k, \phi(\hat{D}_t, \hat{S}_t, \mathcal{R}_{\mathrm{VN}})_k\big). \tag{8}$$

B. Inference Logic for Vietnam

If a valid P.127 sign is detected with confidence above threshold, its value overrides context defaults. Otherwise, inferred speed follows zone sign (R.420/R.421) and hard-divider status. A Bayesian prior update is applied when signs are temporarily occluded but infrastructure evidence is persistent.

VI. Dataset Description

A. Data Sources and Distribution

The dataset is aggregated from uploaded project sources and public sets used for transfer learning. Table I summarizes curated samples used in training/validation/testing.

B. Class Imbalance and Vietnam-specific Factors

In line with project drafts, the class distribution is long-tailed: warning signs dominate while P.127-50 and P.127-80 are scarce. Additionally, the dataset captures Vietnam-specific factors: mixed motorcycle flow, inconsistent sign placement, lighting extremes (night glare, backlight), and camera vibration from low-cost wide-angle installations.

TABLE I
Dataset distribution used in this study.

Source	Train	Val	Test
VNTS (Vietnam signs)	2,420	390	390
Zalo AI Traffic Sign	3,860	613	613
Custom Dashcam VN	3,100	550	550
BDD100K (subset)	4,500	750	750
TT100K (subset)	2,200	360	360
Mapillary (divider subset)	1,800	300	300
Total frames/images	17,880	2,963	2,963

TABLE II
Main hyperparameters.

Hyperparameter	Value
Input size	1280×720
Batch size	16
Optimizer	AdamW
Initial learning rate	2×10^{-4}
Weight decay	1×10^{-4}
Epochs	120
Focal γ	2.0
Focal α	0.25
CIoU weight β	1.0
Dice weight η	0.5
SAHI slice size s	640
SAHI overlap	25%

VII. Model Architecture

The model employs a CSP-based encoder with a feature pyramid neck. The detection head predicts P.127 speed-limit classes and R.420/R.421 zone indicators. The segmentation head predicts lane markings, drivable area, and hard divider masks. For tiny sign recovery, SAHI slicing with 25% overlap is applied at inference and partially during fine-tuning.

If the frame size is $W \times H$, slice size is $s \times s$, overlap is o, then the number of slices is:

$$N_{slice} = \left\lceil \frac{W-o}{s-o} \right\rceil \left\lceil \frac{H-o}{s-o} \right\rceil. \tag{9}$$

Temporal smoothing uses ByteTrack with Kalman filtering to maintain sign identity across brief occlusions.

VIII. Training Strategy

Training runs for 120 epochs with mixed precision and cosine learning rate decay. We use uncertainty-based task weighting to avoid manual tuning of λ_{det} and λ_{seg} under changing scene complexity.

IX. Evaluation and Metrics

Detection performance is reported with Precision, Recall, and mAP. IoU for mask quality is:

$$\mathrm{IoU} = \frac{|M_{pred} \cap M_{gt}|}{|M_{pred} \cup M_{gt}|}. \tag{10}$$

For C classes and IoU thresholds $\tau \in \{0.50, 0.55, \ldots, 0.95\}$, mean AP is:

$$\mathrm{mAP} = \frac{1}{|\tau|}\sum_{\tau}\frac{1}{C}\sum_{c=1}^{C}\mathrm{AP}_{c,\tau}. \tag{11}$$

TABLE III
MODEL COMPARISON ON VIETNAMESE MIXED-TRAFFIC BENCHMARK.

Model	mAP@0.5	mIoU	Lane IoU	FPS
Faster R-CNN (STL)	75.4	–	–	6
YOLOv8s (STL)	77.6	–	–	56
HybridNets	76.8	90.7	27.9	19
YOLOP baseline	76.9	91.6	26.5	24
Proposed (Ours)	**82.6**	**92.3**	**31.9**	**39**

TABLE IV
DETECTION AND SEGMENTATION QUALITY OF THE PROPOSED MODEL.

Target	Precision	Recall	mAP@0.5
P.127 speed signs	0.889	0.832	0.862
R.420/R.421 zone signs	0.901	0.845	0.879
Hard divider (mask)	0.917	0.884	0.903
Lane marking (mask)	0.905	0.861	0.894
Overall weighted	0.903	0.856	0.882

We also evaluate end-to-end speed inference accuracy and stability under occlusion levels 0–3.

X. EXPERIMENTAL RESULTS

A. Comparison with Baselines

Table III shows that the proposed system improves both sign detection and infrastructure segmentation while preserving real-time speed.

B. Precision/Recall/mAP by Task

C. Ablation Study

TABLE V
ABLATION STUDY OF KEY MODULES.

Configuration	mAP small	mAP all	Speed Acc.
Base multi-task model	34.5	77.1	84.0
+ Context rule module	35.2	78.0	89.3
+ SAHI inference	47.1	80.4	90.0
+ SAHI fine-tune	50.6	81.7	90.8
+ Tracking + Bayesian fusion	**51.8**	**82.6**	**92.1**

XI. RESULT ANALYSIS

The strongest gain appears in small-object mAP, confirming that Vietnamese speed signs are frequently too small for plain one-pass detection. SAHI contributes +17.3 points over the base on tiny objects (Table V). Context reasoning contributes the largest gain in final speed-state accuracy because it handles cases where signs are missing or occluded.

Under occlusion level 3 (more than 60% sign coverage), stand-alone TSR recall drops steeply, while lane/divider segmentation remains relatively stable. Temporal-context fusion reduces output flicker and produces smoother speed advisories. Performance degradation is still visible at heavy rain and night glare, especially with low-cost sensors and camera shake.

XII. LIMITATIONS

First, slicing-based inference increases latency, and hardware optimization remains necessary for entry-level embedded platforms. Second, divider semantics can be ambiguous in borderline road markings, occasionally causing 10 km/h inference mismatch. Third, legal reasoning currently relies on predefined rule tables and does not yet parse complex textual supplementary plates.

XIII. FUTURE WORK

Future directions include multimodal fusion with HD maps and GPS for redundancy, robust low-light enhancement for monsoon and night scenarios, and graph-based parsing of compound sign panels in Vietnamese roads. We also plan to add uncertainty-aware human-machine interface feedback so drivers can inspect why a speed value is suggested.

XIV. CONCLUSION

This study demonstrates that ADAS speed assistance in Vietnam requires more than sign recognition. By integrating sign detection, lane/divider segmentation, and legally grounded context inference, the proposed system improves robustness under mixed traffic, non-standard signage, and occlusion-heavy scenes. The experimental results indicate practical potential for localized ADAS deployment and provide a reproducible baseline for future academic and industrial work on Vietnamese intelligent transportation.

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